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PRODUCT REVIEW · HYBRID INVERTER + BATTERY

AlphaESS SMILE-G3-S5

5 kW single-phase hybrid with SMILE-G3-BAT-10.1P LFP battery · pros, cons and market comparison · prepared 9 November 2026 · Solar ESS Working Group

VERDICT

> A tidy, quiet all-in-one with generous PV oversizing and battery expansion — **good for single-phase homes and villas up to about 5 kW**. A weaker fit for three-phase sites, motor loads, whole-house backup and modern high-current panels. **After-sales support in Thailand must be confirmed in writing before we quote it.**

KEY SPECIFICATIONS (MANUAL V01, 2022)

AC rating	5 kW / 5 kVA single-phase, 230 V, 50/60 Hz (S3.6 = 3.68 kW; B5 = AC-coupled, no PV)
Grid pass-through	10 kW rated input, 43.5 A max; 63 A grid breaker
Backup (UPS)	5 kW / 5 kVA, 21.7 A, switchover < 20 ms; 32 A backup breaker; no surge rating stated
PV input	Up to 10 kWp (200%); 580 V max; MPPT 100–550 V; start-up 90 V
MPPT	2 MPPT × 1 string; 15 A operating / 18.75 A short-circuit per MPPT
Battery port	HV Li-ion 80–450 V; 60 A; 5 kW max charge and discharge; CAN
Battery module	SMILE-G3-BAT-10.1P, LFP, 10.1 kWh (9.6 kWh usable, 95% DoD), 96 V, 90 kg, IP65
Expansion	1 to 6 modules per inverter (up to 60.6 kWh)
Battery temperature	Charge 0–50 °C; discharge –10–50 °C (derates with temperature and SOC)
Efficiency	Max > 97%; EU > 96.2%
Enclosure	IP65, natural convection (no fan), < 30 dB(A), –25 to +60 °C, 19.5 kg, 3000 m
Metering / comms	Grid CT (100 A) or Chint DTSU666 meter; LAN standard, WiFi optional; AlphaCloud
Grid codes	Includes PEA and MEA (plus VDE, EN 50549, AS/NZS 4777.2, IEEE 1547 and others)
Off-grid	Off-grid with generator; parallel diagram shows 3 units (1 host + 2 followers)
Warranty (manual V01, 2022)	Inverter 5 years; battery 5 years product / 10 years performance

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PROS AND CONS

Pros

- **200% PV oversizing** (10 kWp on a 5 kW unit) — gains output in mornings, evenings and cloudy weather.
- **One brand for inverter, battery and BMS** — no compatibility disputes on a warranty claim.
- **Large, simple expansion** — up to 6 × 10.1 kWh modules; IP65 so batteries can go outdoors.
- **Fanless and quiet** (< 30 dB) — suits homes, villas and resorts; no fan to fail in dust.
- **PEA and MEA already on the grid-code list** — simpler connection paperwork in Thailand.
- **Off-grid capable** with generator start and up to 3 units in parallel — usable for remote homestays.
- **Newer datasheets show a 10-year product warranty** (the 2022 manual still says 5 years).

Cons for Solar ESS sites

- **Single-phase only.** Three-phase sites need the separate SMILE-G3-T series.
- **Backup capped at 5 kW** on a 32 A breaker, with no surge rating in the manual. Several air-conditioners or a pump starting in a blackout may trip it.
- **10 kWp of PV but only 5 kW AC and 5 kW battery charging.** On zero-export sites the extra solar is curtailed once the battery is full.
- **Tight PV current:** 15 A / 18.75 A Isc and one string per MPPT. Many 600 W+ modules are close to or above this.
- **Heavy battery:** 90 kg per module — 2–3 people or a lifting aid; more labour on hilly or upstairs sites.
- **Battery charges only between 0 and 50 °C** — must be shaded and ventilated in Thailand.
- **Power does not grow with capacity:** 60 kWh of storage still delivers only 5 kW / 60 A.
- **Grid-forming wiring diagrams are marked Australia / New Zealand only** — a Thailand neutral-earth diagram is needed.
- **Thai manual is a machine translation** — not fit to hand to customers as it is.
- **After-sales in Thailand not confirmed:** the 2024 global brochure does not list Thailand among its key Asian service networks.

WHAT IT MEANS FOR THAILAND

Sizing rule

- Size PV to daytime load, not 200%, when the site is zero-export — otherwise the extra solar is curtailed after the battery fills.
- Keep backup loads under 5 kW continuous; avoid large motor starts on the backup port.
- Battery power stays at 5 kW however many modules are added — add modules for hours, not for power.

Installation

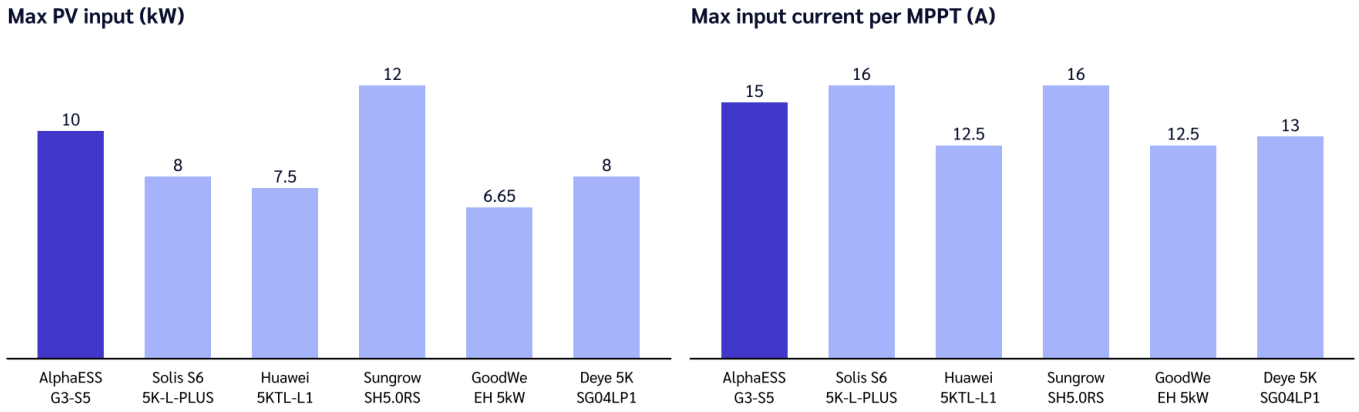
- Mount batteries in shade with airflow; never on a west wall in full sun (charge limit 50 °C).
- Plan lifting for 90 kg modules (up to 6 per system, 300 mm apart side by side).
- Check module Isc against 18.75 A and use one string per MPPT.
- Budget the optional WiFi module, or run LAN.

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COMPARISON WITH SIMILAR 5 KW SINGLE-PHASE HYBRIDS

	AlphaESS SMILE-G3-S5	Solis S6-EH1P5K-L-PLUS	Huawei SUN2000-5KTL-L1	Sungrow SH5.0RS	GoodWe EH 5 kW	Deye SUN-5K-SG04LP1
Battery voltage	**HV 80–450 V**	LV 48 V	HV 350–560 V	HV 80–460 V	HV 85–460 V	LV 40–60 V
Battery choice	**Own brand only**	Open list	LUNA2000 only	Own / listed	Own / listed	Open list
Max PV input	**10 kWp**	8 kW	7.5 kWp	12 kWp	6.65 kW	8 kW*
Current per MPPT	**15 A (1 string)**	16 A (2 strings)	12.5 A	16 A (Isc 20 A)	12.5 A	13 A*
Backup	**5 kW, < 20 ms**	Yes	Via Backup Box	5 kW, < 10 ms	5 kVA	5 kW, 2× peak 10 s
Max efficiency	**> 97%**	—	98.4%	97.7%	97.6%	97.6%
Product warranty	**5 yr (manual) / 10 yr (new sheets)**	10 yr	10 yr	10 yr	10 yr**	5 yr (10 optional)
Service in Thailand	**To be confirmed**	Good back-office support	Direct; replacement delivered to door	To be confirmed	To be confirmed	Weak; via distributors

* Deye figures are for the older SG04LP1 version; newer SM2 versions differ. ** GoodWe: 10 years per current Thai terms; some EU listings show 5 years. “To be confirmed” means we have no verified Thai claim route yet. Specifications from manufacturer datasheets and listings; check the latest datasheet before quoting.



Highlighted: AlphaESS SMILE-G3-S5.

WHERE WE WOULD USE IT

Good fit

- Single-phase villas, homes and homestays wanting a neat outdoor all-in-one system.
- 10–30 kWh storage for mostly evening loads.
- A backup circuit for lights, fridge, Wi-Fi and a small pump.
- Grid-export or self-use sites where PV is sized to daytime load.

Avoid

- Three-phase sites (offer Solis, Huawei or the G3-T series instead).
- Workshops and farms with motor loads; whole-house air-conditioning backup.
- Zero-export roofs with 200% oversized PV.
- Large-format modules unless Isc is checked against 18.75 A.

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CONFIRM IN WRITING BEFORE QUOTING

- Thai product warranty term in writing (5 or 10 years) and who issues it.
- Claim and replacement route in Thailand: named contact, response time, spare stock.
- Isc limit on the hardware revision actually supplied (check the unit label).
- Thailand-specific wiring diagram (neutral / earth, backup) and PEA/MEA approval documents.
- Backup surge rating (kVA for how many seconds).
- A proper Thai-language manual for handover.

SOURCES

1. AlphaESS SMILE-G3 installation manual V01 (2022), EN and TH; SMILE-G3-S wiring diagrams; G3-S5 parallel SLD (supplied files)
2. AlphaESS SMILE-G3 datasheet (solarmagazine.nl); AlphaESS Philippines brochure 2025; AlphaESS Global brochure 2024 (alphaess.com)
3. Solis S6-EH1P5K-L-PLUS listing (sun.store); Huawei SUN2000-L1 specifications (solar.huawei.com)
4. Sungrow SH5.0RS specifications (growenergy.com.au, energymatters.com.au); GoodWe EH datasheets
5. Deye SUN-3-6K-SG04LP1-AU datasheet